Same-drug continuity, not a target profile: a negative result for trial adverse-event transfer

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Abstract

Biological features are often added to chemical fingerprints in the hope that they survive a change of population. On smoothed adverse-event incidence from ClinicalTrials.gov, that hope fails once evaluation is restricted to structures absent from the adult-trial rows. The confirmatory task is adult-only-trial to pediatric-only-trial transfer of a reporting rate, not clinical pediatric toxicity and not an in-vitro assay. A ridge model with pre-specified penalty 10, Morgan fingerprints, and RDKit descriptors has macro Spearman 0.0314 on twelve endpoints frozen before fitting. Early fusion with a 15-target ChEMBL profile (mean missingness 0.9242) reaches 0.0339. The difference is 0.0025 (95% CI -0.0106 to 0.0149). Against five training-row permutations of the profile the difference is 0.0031 (95% CI -0.0143 to 0.0213). A coverage gate differs from chemistry by -0.0007 (95% CI -0.0029 to 0.0014) and raises the negative-transfer count from 5 endpoints to 7. The same chemistry model scores 0.1923 on structures that do appear in both trial sets. The historical appearance of a pediatric signal is largely that overlap. The null is narrow: it concerns this sparse channel and this label.

1 Introduction: expected versus observed

A recurring expectation in molecular machine learning is that a biological representation should help when chemistry alone is asked to move to a new set of compounds. The desired improvement here was specific. On pediatric-only-trial adverse-event incidence, early fusion of chemistry with a ChEMBL target profile, and a gate that trusts that fusion only when the profile is observed, should beat chemistry, and should beat chemistry plus a profile whose compound–target correspondence has been destroyed. Negative transfer, in the sense of a source-trained multimodal score that ranks the target worse than the chemical score [Zhang et al., 2023], should become less common under the gate.

What was observed is a null on all three pre-specified tests (Section 6). The same chemical model looks much stronger on pediatric-trial rows whose structure was already present in an adult trial (macro Spearman 0.1923) than on rows whose structure was not (0.0314). An archived review of an earlier model on these tables already described an overall mean near 0.12 as continuity of the shared structures rather than generalization. This paper's contribution is to freeze the structure-unseen slice, the eligibility rule, and the three contrasts before fitting, and to report the failure with intervals.

The four elements requested by the workshop call are marked in place. **Problem.** Section 3. **Proposed approach.** Section 5. **Observed outcome.** Section 6. **Reason for failure.** Section 7.

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2 Related work

MoleculeNet popularized scaffold-aware evaluation of in-vitro toxicity models and included the Tox21 assay panel [Wu et al., 2018, Tice et al., 2013]. Extended-connectivity fingerprints [Rogers and Hahn, 2010] and Bemis–Murcko frameworks [Bemis and Murcko, 1996] are the chemical representation and the grouping used here. The Chemical Checker showed that bioactivity signatures can extend similarity beyond structure [Duran-Frigola et al., 2020]. Those signatures were not regenerated for this study: the generator did not import, and a data license was not established. An archived Tox21 notebook, using a different model, reported pooled task-macro AUROC 0.8224 for chemistry and 0.8254 for chemistry plus nine signature spaces. The difference of those two saved figures is 0.0030. That gap is prior exploratory context. It is not a result of the fit below, and it is not a trial-incidence result.

ChEMBL is a public bioactivity resource [Zdrazil et al., 2024]. The matrix used here is a 15-target profile inherited from an earlier adult-coverage filter, not a current full ChEMBL release. The release identifier was not in the handoff. Correlation alignment (CORAL) is a transductive covariance-matching method that does not use target labels [Sun et al., 2016]. It is reported only as an exploratory arm. Ridge regression is the confirmatory estimator [Hoerl and Kennard, 1970], fit with scikit-learn [Pedregosa et al., 2011]. Tree models use the pre-specified sizes in the config, including XGBoost [Chen and Guestrin, 2016], and do not choose the claim.

3 Problem setup and task renaming

Problem. The biological task that the files can support is ranking of smoothed logit adverse-event incidence across drug structures. The source domain is adult-only interventional trials (minimum age at least 18). The target domain is pediatric-only trials (maximum age at most 17). Mixed-age trials were already excluded by the upstream aggregate [National Library of Medicine, 2026]. The label is $(a + 0.5)/(n + 1)$ and then the logit, where a is the count affected and n is the count at risk. The loader checks that this recomputation matches the stored column. The quantity is a trial-reporting construct. It is entangled with indication, dose, schedule, and how completely a trial records events. It is not a causal risk and not a developmental-toxicity call.

Calling the target "pediatric toxicity" would overstate the label. Tox21 labels are in-vitro assay activities. FAERS reporting fractions, used in earlier probes of 137 and 46 compounds, do not match either Tox21 or this incidence label, and this paper reports no performance number for them. The renamed confirmatory task is adult-trial to pediatric-trial adverse-event incidence transfer.

The primary population is the pediatric-trial structures whose InChIKey [Heller et al., 2015] is absent from every adult-trial row: 201 structures, against 634 structures present in both sets, out of 835 pediatric-trial and $2{,}856$ adult-trial structures. A pre-fit count found that 48 of those 201 still share a Bemis–Murcko scaffold with an adult structure. Scaffold-unseen rows are a sensitivity slice, not the primary filter.

The target metric is the unweighted mean, across eligible endpoints, of Spearman rank correlation. Accuracy is not used. AUROC is not used on the logit incidence label. A multimodal benefit required both of the following 95% endpoint-bootstrap intervals to lie entirely above 0: early fusion minus chemistry, and early fusion minus the mean of the permutation seeds. A gate gain required the coverage-gate interval versus chemistry to lie entirely above 0. Separately, the gate reduces negative transfer only if it has strictly fewer endpoints with a negative delta than early fusion and its macro Spearman is at least chemistry minus 0.005. Intervals that include 0 are not superiority claims. Benjamini–Hochberg q-values are reported across the three tests [Benjamini and Hochberg, 1995].

4 Data provenance

Chemistry is recomputed from SMILES: Morgan radius 2, $2{,}048$ bits [Rogers and Hahn, 2010], plus 217 RDKit descriptors [RDKit developers, 2026], with a parameter-free $\log(1+x)$ on two unbounded indices before any split. The model matrix has $2{,}265$ chemistry columns after the training-fold filters. Twelve InChIKeys had conflicting SMILES; the mode was kept. Standardization failures were 0 on $3{,}057$ structures. Median imputation, clipping, a variance filter, and scaling are fit on training rows only.

Table 1: Confirmatory contrasts on the structure-unseen slice. Ridge, penalty 10, twelve eligible endpoints. Intervals are percentile bootstraps over endpoints (1,000 draws, seed 0).

Contrast	Delta	95% CI low	95% CI high
Early fusion minus chemistry	0.0025	−0.0106	0.0149
Early fusion minus permuted biology	0.0031	−0.0143	0.0213
Coverage gate minus chemistry	−0.0007	−0.0029	0.0014

The biological channel is 15 ChEMBL targets plus 15 missingness indicators. Mean missingness is 0.9242, mean coverage is 0.0758, and the fraction of structures with any observed target is 0.3811. Coverage is one minus mean target missingness. The profile was not derived from the trial labels. Chemical Checker signatures are absent; treating the code license of the generator as a data license would be a mistake. The ChEMBL-derived matrix is not redistributed with this draft.

Endpoints need at least 200 adult rows and at least 40 structure-unseen rows. That cutoff was fixed before the confirmatory fit. It keeps 12 of 15 endpoints and drops congenital/developmental (24 unseen), hepatic (32), and renal (34). Per-endpoint unseen counts on the kept endpoints run from 49 (cardiac) to 200 (the two aggregate labels).

5 Protocol

Proposed approach. The hypothesis was that, where targets are measured, the profile carries information about adverse-event incidence that fingerprints miss, and that a gate

$$\hat{y} = (1 - g)\,\hat{y}_{\text{chem}} + g\,\hat{y}_{\text{early}},$$

with g equal to the fraction of the 15 targets observed, would fall back to chemistry when the profile is empty and would therefore cut negative transfer.

The confirmatory estimator is ridge regression with penalty 10, solver `lsqr`, not tuned on pediatric labels [Hoerl and Kennard, 1970]. A nested penalty search exists in a full config and was not run. Adult in-domain scores are five-fold GroupKFold on Bemis–Murcko scaffolds. Fusion weights and isotonic maps are fit on those adult out-of-fold predictions only. Permutation seeds are 101, 202, 303, 404, and 505. Each permutation shuffles training rows of the biological block and leaves test rows intact. The endpoint-bootstrap uses 1,000 resamples and seed 0. Random forest and XGBoost are robustness checks at 40 trees. Their adult out-of-fold loop is off in this run. CORAL, after a source-fit principal-component projection of width 32, and importance weighting see unlabeled pediatric chemistry [Sun et al., 2016]. A target-only ridge reference uses labels on half of the structure-unseen compounds. None of those three is a confirmatory test. Adversarial and kernel alignment methods were not fit, because a fair tuning loop would have been tempted to use the transfer labels.

Leakage checks on the fit recorded disjoint adult folds, a permutation that preserves the training multiset, preprocessing statistics that ignore the held-out fold, and a structure-unseen mask with no adult overlap. The wall clock of the confirmatory process was 626 seconds on four threads, with features already cached, covering 15 trial endpoints and a separate Tox21 logistic arm.

6 Observed outcome

On the structure-unseen slice the macro Spearman values are 0.0314 (chemistry), 0.0339 (early fusion), and 0.0308 (coverage gate). Table 1 is the confirmatory family. All three intervals include 0. The Benjamini–Hochberg q-values are 0.7600, 0.7600, and 0.7600. A Wilcoxon signed-rank p-value on the twelve early-minus-chemistry deltas is 0.5186. Endpoints share compounds, so that p-value is descriptive only. The pre-specified multimodal benefit is not supported. The gate does not meet the gain rule or the negative-transfer rule: 7 endpoints versus 5.

Changing the unseen-count cutoff after the fact does not create a positive claim. At a minimum of 20 unseen structures (15 endpoints) the early-minus-chemistry macro delta is 0.0049; at 60 (10 endpoints) it is 0.0019. The gate deltas are −0.0023 and −0.0014. Chemistry's own macro interval on the primary slice includes 0: 0.0314 (95% CI −0.0165 to 0.0828).

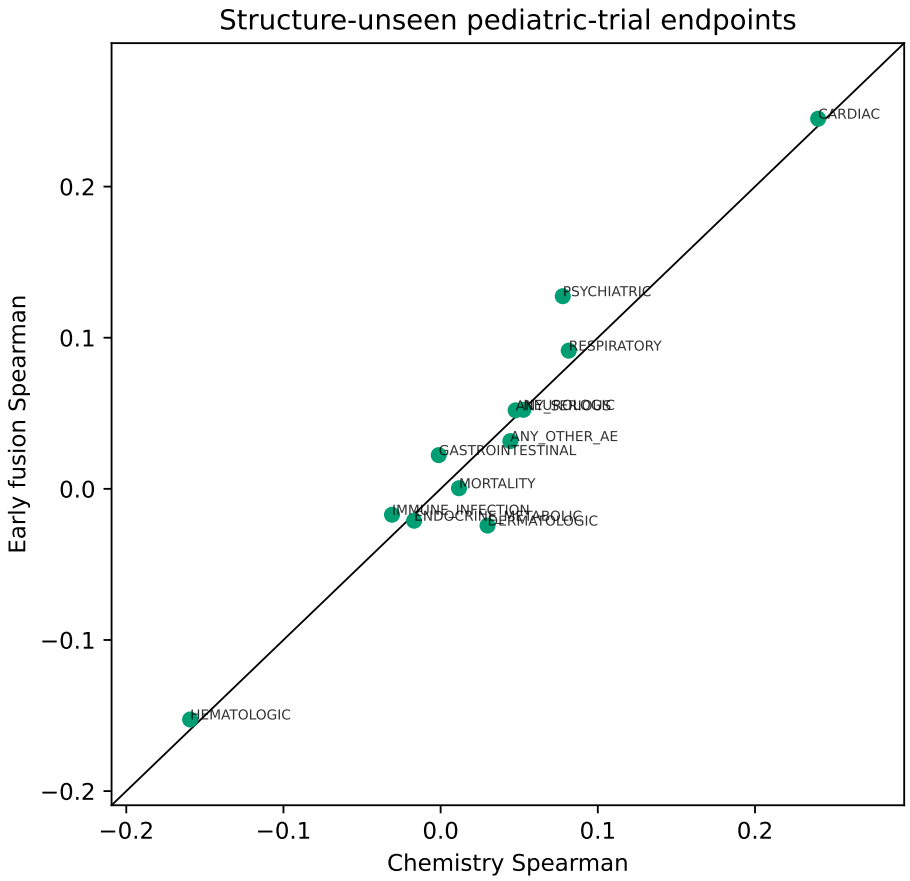


Figure 1: Per-endpoint chemistry versus early fusion on the structure-unseen slice. Points near the diagonal are endpoints where the profile does not move the ranking.

Figure 1 compares the two scores endpoint by endpoint, and Figure 2 shows the paired differences. Dermatologic is the largest negative point estimate (-0.0541, $n = 82$). Psychiatric is the largest positive (0.0498, $n = 64$). The full list is in the appendix. Adult out-of-fold selection set the global mixture weight to 1, which discards chemistry and returns early fusion. There was no in-domain headroom with which to learn a cautious gate. Adult scaffold chemistry on the same twelve endpoints has macro Spearman 0.0220 (95% CI 0.0092 to 0.0386) and macro RMSE 3.0621.

7 Why the expected gain did not appear

Reason for failure. Four facts, all computed from the saved predictions, account for the null without promoting a new primary test.

The apparent signal is same-drug continuity. Table 2 decomposes the pediatric rows. Chemistry's macro Spearman is 0.1923 on structure-seen rows (mean n per endpoint 404.6667), 0.0314 on structure-unseen rows (mean n 109.6667), and 0.1434 on all pediatric rows (mean n 514.3333). Early fusion tracks chemistry on each of those slices (0.1959, 0.0339, 0.1480). A pooled "pediatric" number mostly restates drugs the adult model has already seen. Scaffold-unseen chemistry is 0.0180 (mean n 84.8333), close to the structure-unseen result rather than to the seen-structure result.

The biological channel is mostly missing. Mean missingness 0.9242 means the gate is near zero for most compounds, so the gated prediction is nearly the chemical prediction (delta -0.0007). Where any target is observed, mean n is 24 and chemistry (0.2040) is at least as high as early fusion (0.1831). Where no target is observed, biology's Spearman is undefined because that block does not vary, and chemistry is -0.0093. Missingness indicators alone have macro Spearman 0.0246 (95% CI -0.0381

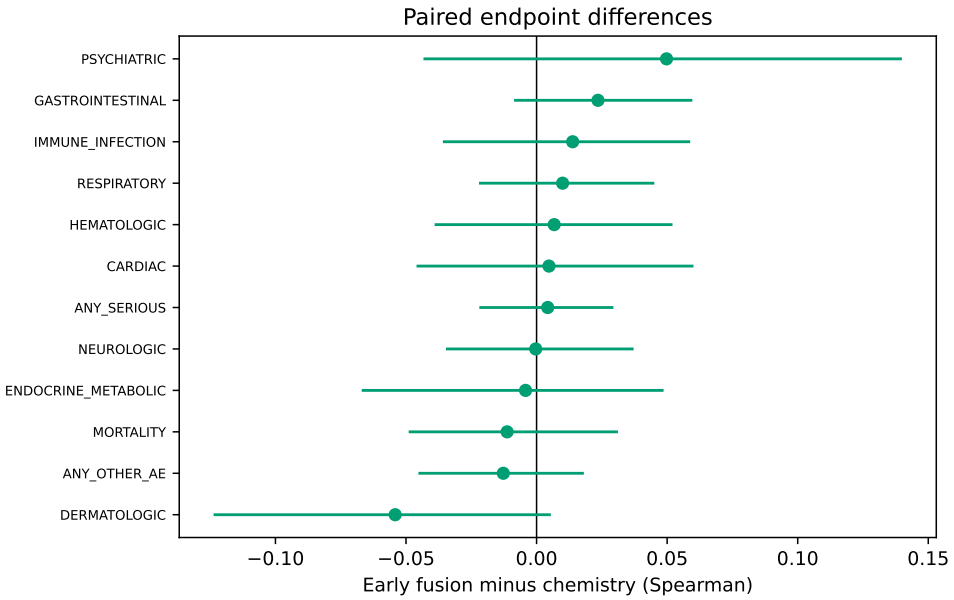


Figure 2: Endpoint-level early-fusion minus chemistry Spearman on the structure-unseen slice. The macro difference is the confirmatory estimand; several endpoints move in opposite directions.

Table 2: Exploratory slice means for ridge Spearman. Structure-seen rows share an InChIKey with the adult trials. Coverage strata split the structure-unseen rows. Blank biology cells are undefined Spearman.

Slice	Chem.	Early	Biology	Mean n
Structure-seen	0.1923	0.1959	0.0837	404.67
Structure-unseen	0.0314	0.0339	0.0723	109.67
All pediatric	0.1434	0.1480	0.0766	514.33
Unseen, any target	0.2040	0.1831	0.1843	24.00
Unseen, no target	−0.0093	−0.0081	—	85.67

to 0.0821), so the indicators are not a hidden label leak. Figure 3 shows how thin the observed profile is.

The chemical baseline is weak on the shift that matters. Adult scaffold Spearman 0.0220 is detectably positive and practically small, and the structure-unseen chemistry interval includes 0. A fusion test on top of a near-null ranking has little headroom. The adult macro RMSE of 3.0621 on the logit scale is a second sign that the pre-specified penalty does not produce a sharp chemical score. Because the penalty was not chosen on adult folds, this paper cannot say whether a different frozen penalty would move the point estimate. It can say that the penalty was not chosen on the transfer test.

The learned weight collapsed. The adult out-of-fold grid selected weight 1 on early fusion. Early and chemistry differ by 0.0004 Spearman in that domain (0.0224 versus 0.0220). A validation rule with no source-domain gap cannot learn to ignore an unreliable channel. Fixed 50/50 mixtures and endpoint-wise weights land in the same narrow band as early fusion (macro Spearman 0.0330 to 0.0345 on the unseen slice). Random and permuted gates do as well (0.0307 and 0.0323). The non-deployable oracle, which looks at the label, reaches 0.0690 and is only a ceiling.

Boundary conditions follow directly. Do not read a multimodal null on a 15-target, 92% missing profile as a statement about biological representations in general. Do not read a structure-seen Spearman near 0.19 as evidence that the model generalized to new drugs. Do not deploy the score as a clinical or regulatory safety tool.

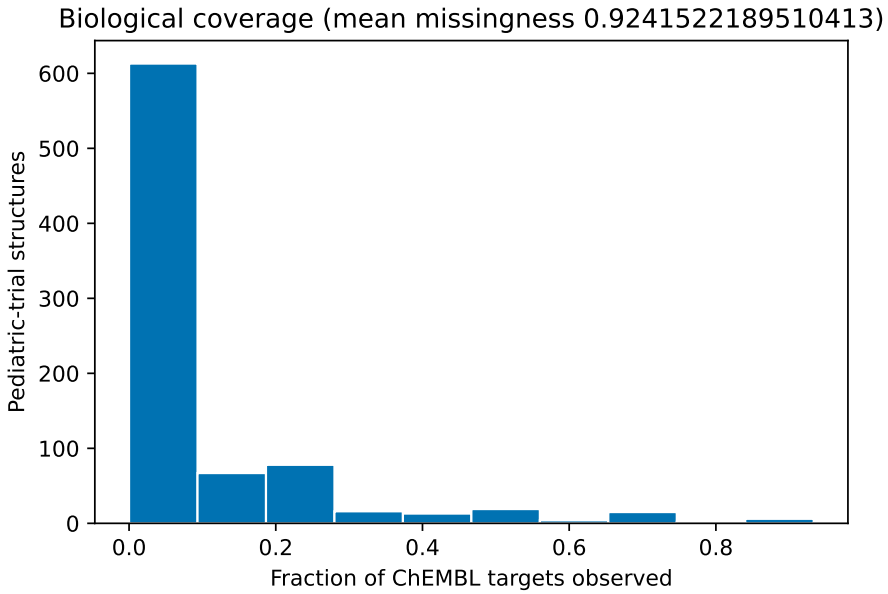


Figure 3: Biological coverage on the trial structures. Most compounds have no observed ChEMBL target in this 15-target vocabulary, so a coverage gate has little room to depart from chemistry.

8 Lessons for multimodal biomedical transfer

Overlap audits come before transfer claims. Here, 634 of 835 pediatric-trial structures are already in the adult table. Removing them changes the chemical Spearman from 0.1434 to 0.0314. Any benchmark that scores a "new population" without an identifier-level exclusion will credit continuity.

A reliability gate needs a source-domain signal to train against. Weight collapse to pure fusion is a failure mode of validation-set mixing when the two source scores are interchangeable. The gate then cannot protect the target.

Report the channel, not the word "biology." Coverage, the fraction of compounds with any observation, and a missingness-only control belong next to the multimodal delta. In this file those three numbers are 0.0758, 0.3811, and a contrast versus chemistry whose interval includes 0 (missingness minus chemistry -0.0068, 95% CI -0.0747 to 0.0739).

Pre-register the family of tests. Exploratory level intervals that exclude 0 are easy to promote after the fact. Biology-only and CORAL are in that category and are confined to the appendix. They do not overturn Table 1.

Name the label. Trial incidence, spontaneous-report fractions, and assay hits answer different questions. Mixing them produces an endpoint shift that no representation can honestly paper over.

9 Limitations

The structure-unseen sets are small, and the twelve endpoints are not independent. The endpoint bootstrap matches the claim and is still a bootstrap of twelve numbers. Forty-eight of the 201 unseen structures share a scaffold with the source, so even the primary slice is not a pure scaffold holdout. The ChEMBL release is unidentified, and the vocabulary was chosen by an earlier adult-coverage filter. Chemical Checker signatures were not built. The full nested-penalty config was not executed. CORAL and importance weighting see unlabeled target chemistry; the target-only reference sees target labels. Indication and reporting completeness remain inside the label. Nothing here is a clinical decision rule.

Broader impact

The useful outcome is a negative result with the endpoint named correctly. Publishing structure-seen performance as pediatric generalization would overstate what a trial-reporting model can do for dosing, developmental toxicity, or regulatory screening. The trial aggregates are structure-level counts from a public registry, not patient records. A ChEMBL-derived matrix carries a share-alike obligation and an unidentified release, so it is not included with this draft. The predictions required to recompute the decimals are structure identifiers, endpoint names, smoothed labels, and scores.

Reproducibility

The confirmatory command is `python scripts/reproduce_all.py -config configs/reduced.yaml`. It refuses to start unless the analysis plan contains the frozen-protocol marker, which was committed before the fit. The fit manifest records commit `a9324cf`, timestamp 2026-10-19T15:29:45Z, penalty mode prespecified, and the package versions used (Python 3.12.3, scikit-learn 1.9.1, RDKit 2026.9.1, XGBoost 3.4.1). The trial prediction file has SHA-256 `37dbf42edac39337d9e76aeb446661dd7e3a0e16429d453cfd4d93abf68faa0d`. Tables and figures are functions of that file. The synthetic smoke config does not use the trial tables and does not update the claim. The nested-penalty command was not the source of any number in Table 1. Linked code should be stripped of repository URLs and names before review, because the public archive identifies the authors.

Use of large language models

A large language model drafted the prose of this manuscript from the saved tables and the frozen analysis plan. It did not choose the eligibility cutoff, the penalty, the permutation seeds, or the three confirmatory tests. Numerical sentences were checked against those tables. A human author must still read the compiled PDF before any submission.

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A Exploratory arms that are not confirmatory

Table 3 repeats endpoint-bootstrap intervals that were not in the Benjamini–Hochberg family. Biology-only macro Spearman is 0.0723 (95% CI 0.0042 to 0.1441). Biology minus chemistry is 0.0409 (95% CI -0.0383 to 0.1220). The second interval includes 0. CORAL chemistry is 0.0985 (95% CI 0.0172 to 0.1753), and CORAL minus chemistry is 0.0671 (95% CI -0.0588 to 0.1711). CORAL sees unlabeled pediatric features [Sun et al., 2016]. Importance-weighted chemistry is 0.0239 (95% CI -0.0154 to 0.0639). Early fusion without missingness indicators differs from chemistry by 0.0025 (95% CI -0.0041 to 0.0100). A chemistry model trained on permuted adult labels scores -0.0348, which is a negative control rather than a transfer method.

Table 3: Exploratory endpoint-bootstrap intervals on the twelve eligible endpoints. Not adjusted with the three confirmatory tests.

Quantity	Estimate	95% CI
Biology only	0.0723	$0.0042, 0.1441$
Biology minus chemistry	0.0409	$-0.0383, 0.1220$
CORAL chemistry	0.0985	$0.0172, 0.1753$
CORAL minus chemistry	0.0671	$-0.0588, 0.1711$
Importance-weighted chemistry	0.0239	$-0.0154, 0.0639$
Missingness only	0.0246	$-0.0381, 0.0821$

On a scaffold holdout that uses target labels for half of the structure-unseen compounds, ridge chemistry has macro Spearman 0.1211 and early fusion 0.1166. That arm is a labeled reference, not a zero-shot transfer, and fusion does not win it.

Tree point estimates on the same unseen slice, without a pre-specified interval, are: random-forest chemistry 0.0548 and early fusion 0.0244; XGBoost chemistry 0.0050 and early fusion 0.0552. The XGBoost gap was not a confirmatory test, adult out-of-fold predictions were not used to set its hyperparameters, and no interval was pre-registered. It is reported so that it is not hidden. It is not a claim that gradient boosting plus the profile succeeds.

Moderator correlations of the early-minus-chemistry delta with coverage (-0.5324), scaffold novelty (0.5464), adult chemistry Spearman (-0.1119), unseen n (-0.2312), and mean label (0.1748) use twelve endpoints. They are not causal effects.

Figure 4 compares true and permuted biological features. The permutation seeds' chemistry-plus-biology macro Spearman values on the unseen slice are 0.0278, 0.0284, 0.0470, 0.0212, and 0.0296 for seeds 101, 202, 303, 404, and 505.

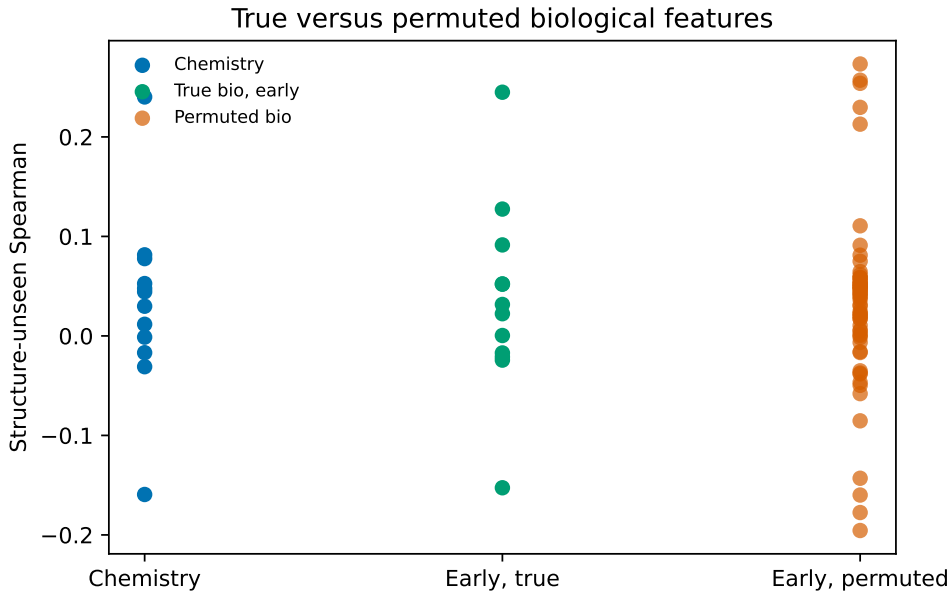


Figure 4: True biological features versus training-row permutations. The confirmatory comparison is early fusion against the mean of these controls, not against a single seed.

B Tox21 chemical baseline

Tox21 is a separate in-vitro question [Wu et al., 2018, Tice et al., 2013]. The arm in this repository is logistic regression on Morgan fingerprints only, with scaffold GroupKFold and a shuffled K-fold, both with five splits. It is not the archived XGBoost pipeline. Biological signatures were not run. Compounds: 7,587. Labels: 75,521. An assay is scored only with at least 20 positives and 20 negatives; all twelve tasks met that bar.

Scaffold-split macro AUROC is 0.6895 and macro AUPRC is 0.2440. Nuclear-receptor assays are 0.7034 and stress-response assays are 0.6700 under that split. The random split, which allows scaffold leakage, has macro AUROC 0.7331 and macro AUPRC 0.3200. The random-minus-scaffold gap is 0.0436. Nuclear receptor versus stress response is a comparison of assay families under one split, not a label-aligned transfer. The gap shows that this chemical baseline does rank assay hits, and that a random split overstates that ranking. It does not rescue the trial-incidence null.

The archived notebook number 0.8224 versus 0.8254 used XGBoost and, for the second figure, Chemical Checker spaces that this fit did not rebuild. Those two AUROCs are not recomputed here.

C Endpoint deltas

Table 4 lists every eligible endpoint. Signs disagree. No endpoint was dropped after seeing its delta. The three endpoints below the frozen count cutoff stay out of the mean.

D Design schematic

Figure 5 is the analysis flow used to generate the predictions. It does not add a numeric claim.

Table 4: Structure-unseen ridge Spearman by endpoint, early fusion minus chemistry. Sorted by the delta. All twelve rows enter the confirmatory macro.

Endpoint	n	Chem.	Early	Delta
Dermatologic	82	0.0298	−0.0243	−0.0541
Any other adverse event	200	0.0443	0.0316	−0.0127
Mortality	144	0.0117	0.0004	−0.0113
Endocrine/metabolic	72	−0.0169	−0.0211	−0.0042
Neurologic	106	0.0526	0.0523	−0.0003
Any serious	200	0.0477	0.0520	0.0043
Cardiac	49	0.2402	0.2449	0.0047
Hematologic	59	−0.1594	−0.1526	0.0068
Respiratory	107	0.0815	0.0915	0.0100
Immune/infection	113	−0.0309	−0.0171	0.0138
Gastrointestinal	120	−0.0012	0.0223	0.0235
Psychiatric	64	0.0777	0.1275	0.0498

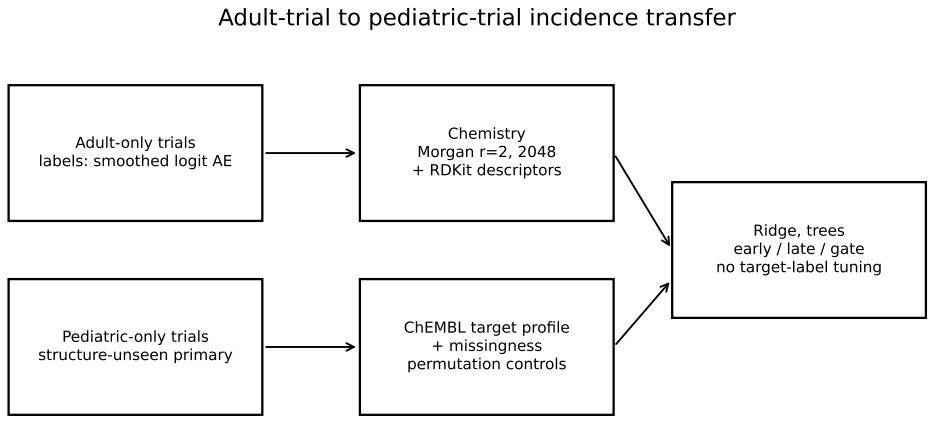


Figure 5: Flow from adult-trial fitting to the structure-unseen evaluation. Counts and metrics in the text are taken from the saved tables, not from this drawing.